UNIVERSITY WEST

# A human-centric framework for next-generation Plug & Produce manufacturing

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ACADEMIC DISSERTATION

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Increasing demand for rapid product variation and customization requires automation that can be reconfigured efficiently without extensive reliance on external expertise. Part-specific information is typically embedded in application programs of traditional control systems, making adaptation to new products complex and time-consuming. The contributions of this thesis are a human-centric, configurable, multi-agent control system at the machine level for next-generation Plug & Produce. The agent's intelligence is enhanced by utilizing automated AI planning for transfer tasks. Part-dependent data are relocated from control logic to product configurations and are performed from the part and process perspectives rather than from programming resource functionalities. Manufacturers extract part data from CAD models and incorporate process knowledge via graphical user interfaces, enabling them to configure systems for new products without specialized programming skills. An approach that opens the door for creative operators to handle the next-generation, human-centric Plug & Produce system. Integrating agents into robot control systems and other external control systems via device primitives makes these systems flexible and independent of part-specific data. A new part agent strategy allows concurrent resources to be plugged in to balance capacity. Low-level controls imply high real-time demands, which are met through short negotiation processes via abstract interfaces that match real interfaces on resource agents, efficient agent communication, and fast automated AI planners.